MIL-T-40001E
29 July 1988
SUPERSEDING
MIL-T-40001D
30 December 1971

MILITARY SPECIFICATION

TENT, OBSERVING, TRIANGULATION, GROUND TYPE, COMPLETE WITH FRAME

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a hexagonal tent with the following dimensions: length of sides, 4 feet; eave height, 6 feet; height at peak, 7 feet 5 inches. The tent is composed of a main wall and a top and drop wall. The tent is supported by a frame consisting of six, sectional, aluminum poles joined to an aluminum spider. A cover is provided to protect the tent and its components during transportation.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8340

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SPECIFICATIONS

FEDERAL

	V-T-285	- Thread, Polyester
	QQ-A-225/8	- Aluminum Alloy 6061 Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished
	QQ-C-390	- Copper Alloy Castings (Including Cast Bar)
	QQ-C-465	- Copper-aluminum Alloys (Aluminum Bronze) (Copper Alloy Numbers 606, 614, 630 and 642); Rod, Flat Products with Finished Edges (Flat Wire, Strip, and Bar), Shapes, and Forgings
	QQ-W-321	- Wire, Copper Alloy
	QQ-W-461	- Wire, Steel, Carbon (Round, Bare, and Coated)
	TT-E-529	- Enamel, Alkyd, Semi-Gloss
*	TT-P-595	- Preservative Coating, Canvas
*	TT-P-1757	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
	WW-T-700/6	- Tube, Aluminum Alloy, Drawn, Seamless, 6061
	DDD-L-20	- Label: for Clothing, Equipage, and Tentage, (General Use)

MILITARY

	MIL-C-496	- Clips, End, Strap
	MIL-W-530	- Webbing, Textile, Cotton, General Purpose, Natural or in Colors
	MIL-L-1709	- Lines, Tent
	MIL-S-1734	- Slips, Tent Line
	MIL-C-2399	- Cement, Liquid, Tent Patching
	MIL-C-5541	- Chemical Conversion Coatings on Aluminum Alloys
	MIL-L-11075	- Loops, Strap Fastener
	MIL-G-16491	- Grommet, Metallic
	MIL-T-43566	- Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
	MIL-C-43627	- Cloth, Duck, Cotton, Plied Yarns, Fire, Water, Weather and Mildew Resistant Treated, Lightdry Finish
*	MIL-W-44049	- Webbing, Textile and Tape, Textile, Polypropylene, General Purpose, Natural or in Colors
*	MIL-C-52950	- Crates, Wood, Open and Covered

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STANDARDS

FEDERAL

FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
AN-960 - Washer, Flat

* (Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

* Department of the Army Field Manual FM 10-16
General Repair of Tents, Canvas and Webbing

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402

DRAWINGS

* US ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

5-4-466 - Tent, Observing, Triangulation, Ground Type, Complete with Frame; Assembly and Details
5-4-467 - Wall and Top Panel Details
5-4-468 - Frame
5-4-469 - Tent, Observing (Astronomic, Ground Type, Magnetic or Tower Type); Cover Assembly and Sections
5-4-470 - Cover Details
5-4-471 - Tent, Observing, Triangulation, Ground Type, Complete with Frame; Erection Instructions
5-4-487 - Packaging Procedure

* (Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of solicitation.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

* 3.2 Materials and components. Materials and components shall be as specified herein (see 6.5).

3.2.1 Cloth.

* 3.2.1.2 Duck, cotton. The cloth for the tent, peak reinforcement and tent cover shall conform to MIL-C-43627 and the color shall be Olive Drab No. 7.

* 3.2.2 Tape, textile, cotton. The tape shall be shade Olive Drab No. 7 conforming to type I, class 4, 1/2 inch, 1 inch and 1-1/2 inch widths of MIL-T-43566. For cover only, alternatively, the tape shall be polypropylene, conforming to type I, class 4, Olive Drab No. 7 of MIL-W-44049.

* 3.2.3 Webbing, textile, cotton. The webbing shall be shade Olive Drab No. 7 conforming to type IIb, class 4, 1 inch width of MIL-W-530. For cover only, alternatively, the webbing shall be polypropylene, conforming to type IIb, class 4, Olive Drab No. 7 of MIL-W-44049.

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3.2.4 Thread, polyester. The thread shall conform to type I, class 1, subclass B, shade S-1 (cable no. 66022) of V-T-285. The dyed thread shall show fastness to weathering equal to or better than the standard shade sample (see 6.3). When no standard sample is available, the dyed thread shall show good fastness to weathering.

* 3.2.4.1 Sizes. Thread sizes shall be as follows:

* For all stitching except stitching indicated by "Z" on drawings and bartacking.	Size F for needle and E for bobbin.
For bartacking.	Size E for needle and bobbin.
For stitching indicated by "Z" on drawings.	Size F for needle, size E for looper.

* 3.2.5 Lines, tent. The tent lines shall conform to types I and XXI of MIL-L-1709. As an alternate, either sisal, polyester or three strand, dual fiber polyester/polypropylene commercial rope may be used. When either of the alternate ropes is used, all other requirements of MIL-L-1709 are applicable.

3.2.6 Clips, end. The end clips shall conform to type I, class 1, size 1-inch of MIL-C-496.

3.2.7 Grommets, metallic. The grommets shall conform to type III, class 3, size no. 4 of MIL-G-16491.

3.2.8 Loops, strap fasteners. The loops shall conform to type I, or II, style A, or type III, style B of MIL-L-11075.

3.2.9 Slips, tent line. The tent line slips shall conform to type II of MIL-S-1734.

3.2.10 Aluminum.

3.2.10.1 Sleeve. The sleeve shall be alloy 6061-T6, conforming to QQ-A-225/8.

3.2.10.2 Tubing. The tubing shall be alloy 6061-T6, type I conforming to WW-T-700/6.

3.2.11 Brass.

3.2.11.1 Escutcheon pins. The wire for escutcheon pins shall be copper alloy No. 260 or 270, half hard temper conforming to QQ-W-321.

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3.2.11.2 Washers, free floating. The washers for spider ring shall be brass conforming to dash number AN 960-B516 of AN-960.

3.2.12 Wire, phosphor-bronze, spring. The wire for helical springs shall be copper alloy No. 510, spring temper, conforming to QQ-W-321.

3.2.13 Wire, steel. The steel wire for formation of spider ring shall be grade SN-1015, finish 1 conforming to QQ-W-461.

3.2.14 Castings, aluminum bronze. The castings shall conform to copper alloy No. G6 or G7, type I, condition as cast, of QQ-C-390.

3.2.15 Aluminum bronze, bar and rod stock. The bar and rod stock and washers shall be aluminum bronze conforming to copper alloy No. 630 or 642, drawn, of QQ-C-465.

* 3.2.16 Primer. The zinc chromate primer shall conform to composition G or L, as applicable, color Y of TT-P-1757.

3.2.17 Enamel. The enamel shall be Olive Drab No. 24087 conforming to class A or B of TT-E-529.

3.3 Construction. The construction shall conform to the drawings listed in section 2 and as specified herein.

3.3.1 Stitching machine.

* 3.3.1.1 Types of stitching. All stitch types, except bartacking, shall conform to FED-STD-751 as follows:

For all stitching except stitching indicated by "Z" on drawings.	Type 301, 5 to 7 stitches per inch.
For stitching indicated by "Z" on drawings.	Type 401, 5 to 7 stitches per inch. Chain portion of stitching shall not appear on outside of cover.
* For bartacking.	1/2 ± 1/16 inch length, 1/8 ± 1/16 inch height (bight) with 28 stitches per inch. 3/4 ± 1/16 inch length, 1/8 ± 1/16 inch height (bight) with 42 stitches per inch.

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* 3.3.1.2 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/2 inch minimum except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in puckering of material sewn. The lock shall be embedded in the material sewn.

* 3.3.1.2.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1 inch behind the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch before the defective area (1/2 inch on box, box and cross and W-W stitching) and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

* 3.3.1.3 Type 401 stitching. Thread tension shall be maintained so that there will be no loose stitching. Both ends of seams or stitching produced with a 401 stitch type, when not turned under a hem or held down by other stitching, shall have a 3/4 to one inch chain extending beyond each end. The looper (under thread) shall always be on the inside of the tent or cover. All repairs shall be made using a 301 stitch in accordance with 3.3.1.2.1a or 3.3.1.2.1b.

* 3.3.1.4 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements of the stitch pattern, stitches per inch, size and type of thread are met, and at least three or more tying, overlapping or backstitches are used to secure the ends of the stitching.

* 3.3.1.5 Thread ends. All thread ends that are visible on the finished item shall be trimmed to a length of not more than 1/4 inch.

* 3.3.1.6 Lubrication of thread. The addition of any lubricant to the polyester sewing thread prior to or during the sewing operation is prohibited.

* 3.3.1.7 Stitching margins. Unless otherwise specified, all stitch margins shall be 1/8, +1/16 -0 inch.

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* 3.3.1.8 Bartacking. Bartacking shall be free of thread breaks and loose stitching.

* 3.3.2 Replacement of defective components. During the spreading, cutting and manufacturing process, components of the tent having material defects or damages that are classified as defects in 4.4.3 or 4.4.4 shall be removed from production and replaced with non-defective and properly matched components.

* 3.3.3 Fusing of ends of synthetic rope, webbing, and tape. All ends of synthetic rope, webbing, and tape shall be fused. The apparatus used to fuse the ends shall be capable of providing sufficient heat to provide a smooth edge and with the cut ends of the yarns all fused together. Fusing of the ends shall be accomplished prior to being assembled for stitching.

* 3.3.4 Splicing of cloth. There shall be no splicing of roof, wall or tent cover panels except as indicated in the drawings. Splicing of sod cloth sections will be permitted and if performed, shall conform to the following requirements:

a. Six feet or less in length - One splice permitted.

b. The shortest length of any spliced piece shall be not less than 2 feet.

c. Pieces shall be joined as shown on Drawing 5-4-467, Section D-D.

* 3.3.5 Splicing of webbing. No splicing of webbing will be allowed on the roof portion. On eave and wall panels, splices will be permitted and if perfomed, shall be in accordance with View A on Drawing 5-4-467. There shall be a minimum of 4 feet between splices and a maximum of five splices per main wall.

* 3.3.6 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. Grommets shall be securely clinched without cutting the materials. Roll of the clinched grommet shall not contain splits.

3.3.7 Setting of end clips. The webbing shall be inserted the full depth of the clip. Clips shall be securely attached without cutting the webbing and the webbing shall be flat at the point of entrance.

3.3.8 Frame. The frame shall conform to the requirements as shown on Drawing 5-4-468.

3.3.9 Welding and brazing. All welding and brazing shall be as specified on the drawings. Materials to be welded shall be free of scale, corrosion, oil, water and other matter. Welding and brazing shall be sound, smooth and clean.

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3.3.10 Finish. All aluminum components shall be prepared for finishing in accordance with class 1a requirements of MIL-C-5541, and given one coat of primer specified in 3.2.16 followed by one coat of enamel specified in 3.2.17. All steel components shall be cleaned of grease or any foreign matter and when dry shall be given one coat of primer specified in 3.2.16 followed by one coat of enamel specified in 3.2.17. The finish shall level out to produce a smooth, uniform surface.

* 3.3.11 Repairs. Repairs, whenever necessary shall be effected in accordance with Department of the Army Field Manual FM 10-16, except that hand-sewn patches, or cemented patches other than specified herein, will not be acceptable. Holes, cuts or tears not more than 1 inch in diameter or length may be repaired with cemented patches. The cement shall be as specified in MIL-C-2399. Three cemented patches per tent will be acceptable without prior authorization. All other repairs shall require contracting officer authorization. Cemented patches will be placed on the outer surface of the tent. The size of the patches shall be such that they will overlap the damage with a margin of at least 3/4 inch on all sides. Repairs required as a result of failures of the cup test shall be performed by treating with a compound conforming to type I of class 1 of TT-P-595 prior to retesting (see 6.7).

* 3.3.12 Wicking of sewing thread (cup test). There shall be no seepage of water through the tent stitching when the tent is tested as specified in 4.4.5. The cut test is intended to insure that non-wicking thread has been utilized and that no lubrication has been added to the polyester thread prior to or during the sewing operation.

3.4 Markings. All labels and markings shall conform to DDD-L-20.

3.4.1 Nomenclature. The letters "U.S." on the tent and cover, and the identification marking on the cover shall conform to type IV, class 9 in the size characters and locations shown on the drawings. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

* 3.4.2 Identification label. Identification labels shall be stitched to the main wall, top and drop wall, and cover in the locations shown on the drawings, and shall conform to type VI, class 6. The size of the label shall be 3-1/2 inches by 5-1/2 inches minimum, with a blank space of at least 1-1/4 inches provided on the bottom of the label for the Department of Defense inspector's stamp.

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* 3.4.3 Erection instruction label. Labels of erection instructions shall conform to type VI, class 7 and shall be stitched to the main wall and cover in the locations shown on the drawings. The label shall be a proportionate copy of Drawing 5-4-471 as reproduced from a glossy print of this drawing furnished by the contracting officer. The size of the label shall be borderline to borderline, 15 ± 1/4 inches in the longer dimension with a minimum 1/2 inch margin outside of the borderline on all sides.

* 3.5 Workmanship. The tent shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

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* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.4.5. The presence of any defect shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 In-process inspection. Inspection shall be made of the following operations or assemblies to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

a. Polyester thread has not been lubricated prior to or during the sewing operations in compliance with 3.3.1.6.

b. Holes punched to receive grommets are in compliance with 3.3.6.

c. Surfaces to be welded are clean and free from scale, corrosion, oil or other matter.

d. Aluminum parts are pretreated, primed, and enameled in compliance with 3.3.10.

e. Steel components and hardware are degreased, primed, and enameled in compliance with 3.3.10.

* 4.4.3 End item visual examination.

* 4.4.3.1 Tent, observing, triangular. The end items shall be in an erected position so that both the inside and outside of the tent can be examined for defects listed in table I. The lot size shall be expressed in units of tents. The sample unit shall be one tent. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 25 for major defects, 100 for major and minor A combined defects and 150 for total (major, minor A, and minor B combined) defects.

* 4.4.3.2 Cover, only. The end items shall be in a position so that both the inside and outside of the cover can be examined for the applicable defects listed in table I. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 10 for major defects, 25 for major and minor A combined defects, and 100 for total (major, minor A, and minor B combined) defects.

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* 4.4.3.3 Frame only. The end item shall be in an erected position so that the frame can be examined for defects listed in table II. The lot size shall be expressed in units of frames. The sample unit shall be one frame. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects, 6.5 for major and minor A combined defects, and 15 for total (major, minor A and minor B combined) defects.

TABLE I. End item visual defects (tent or cover)

Examine	Defect	Classification Major	Minor A	Minor B
Fabric	Cut, tear, or hole:			
	- greater than 1/8 inch in length or diameter:			
	- on tent	X		
	- on cover		X	
	- more than five holes less than 1/8 inch in length or diameter:			
	- on roof		X	
	- walls or cover			X
	Broken or missing yarn; multiple floats, clearly visible at normal inspection distance (approximately 3 feet)		X	
Tape and webbing	Edge frayed or scalloped		X	
Hardware (tent and cover only)	Broken	X		
	Malformed:			
	- fails to perform intended function	X		
	- but will perform intended function		X	
	Corroded area			X
	Burr or sharp edge which may cause injury in handling or damage to fabric	X		
Grommets	Improperly set; two or more teeth exposed			X
	Clinched loosely, allowing grommet to rotate around hole			
	Clinched excessively tight, cutting fabric	X		
	Insecurely clinched; i.e., grommets or washers become disengaged	X		

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TABLE I. End item visual defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
* Grommets (cont'd)	Roll of grommet contains one or splits:			
	- on tent	X		
	- on cover			X
End clips	Not securely clinched; webbing not flat at point of insertion			X
	Clinched excessively tight, cutting webbing		X	
Rope components	Any cut; any break:			
	- on eave guy lines		X	
	Eye splice omitted		X	
	Eye splice not as specified			X
	Free ends not finished as specified			X
Seams and stitching	Open seams on stitch type 301:			
	- more than 1/4 inch but not more than 1 inch			X
	- more than 1 inch		X	
	Open seam on stitch type 401:			
	- one row of stitching open for 1 inch or less:			
	- tent and cover			X
	- one row of stitching open for more than 1 inch:			
	- on roof		X	
	- on wall and cover			X
	- both rows of stitching open (not necessarily at same location):			
	- on tent	X		
	- on cover		X	

NOTE: A seam shall be classified as open when stitches joining a seam are broken, or when consecutive skipped or run-off stitches occur. On double-stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.

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TABLE I. End item visual defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)	Raw edges caught in stitching:			
	- more than 1 inch but not more than 3 inches			X
	- more than 3 inches		X	
	Thread breaks:			
	- overstitched less than 1 inch on stitch type 301			X
	- overstitched less than 1 inch on stitch type 401			X
	NOTE: Thread breaks not overstitched shall be classified as open seams.			
	Needle chews resulting in cut or tear larger than 1/8 inch in length, or hole larger than 1/8 inch diameter (see cut, tear, or hole)			
	One or more required rows of stitching omitted (except on boxstitching)	X		
Boxstitching	Boxstitching incomplete:			
	- one row of stitching omitted		X	
	- two or more rows of stitching omitted	X		
	Any boxstitching less than specified length by more than 1/8 inch			X
"W" stitching	"W" stitching less than specified length by more than 1 inch			X
Bartacking	Bartacking less than specified length by more than 1/8 inch			X
Stitching ends	Overstitched less than 1/2 inch on stitch type 301 (except where ends are held down by other stitching, turned under in a hem, or where stitching is performed automatically)		X	
	Secured with less than 3 tying, overlapping or backstitches when automatic stitching is performed			X

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TABLE I. End item visual defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seam type	Wrong seam type	X		
Stitch type	Wrong stitch type	X		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches		X	
	Tight, as evidenced by puckering on seams - for more than 6 inches		X	
Stitches per inch	Less than minimum specified (except panel joining seams) - one stitch			X
	- two or more stitches		X	
	Less than minimum specified on panel joining seams		X	
	More than maximum specified: - one stitch			X
	- two or more stitches		X	
	NOTE: Variation in the number of stitches per inch, caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or turning corners, shall be classified as follows: a - Within the minor "A" defect classification-minor "B" defect b - Within the minor "B" defect classification-no defect			
Stitching gage	Not as specified		X	

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TABLE I. End item visual defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
* Stitching margin	Larger than specified:			
	- Less than 4 inches in length			X
	- 4 inches or more in length		X	
	Less than specified:			
	- Less than 4 inches in length			X
	- 4 inches or more in length		X	
Stitching indicated by "Z" on drawings	Chain portion of stitching shows on outside of tent or cover		X	
* Mends, darns or patches	Repairs not in accordance with Department of the Army Field Manual FM 10-16; not authorized by the contracting officer (see 3.3.11)	X		
Components and assembly	Any component part not fabricated of applicable referenced material	X		
	Any component omitted	X		
Construction details:				
Hems	Constructed with partial turnunder when double turnunder is required		X	
	Twisted or pleated			X
* Cloth splicing	Splice in roof, wall, or tent cover panels	X		
	Number of splices per sod cloth section more than specified			X
	Spliced piece less than two feet in length			X
	Spliced piece not joined as shown on Drawing 5-4-467, section D-D		X	
* Splicing of webbing	Splicing (when permitted) not made as indicated on Drawing 5-4-467		X	
	Distance between splices more than or less than stated		X	
	Splicing of webbing in roof portion	X		

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TABLE I. End item visual defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Reinforcements	Improperly applied, causing excessive fullness on reinforcements or reinforced parts		X	
Main wall:				
Sod cloths	One or more sod cloths improperly applied in wall bottom hem		X	
Loops	Both ends of loops not securely caught in bartacks		X	
Cover:				
Side flaps	Attached on outside of body in lieu of inside		X	
Pocket flaps	Inner edge not securely caught in stitching attaching strap "A" to body			X
Labels	Omitted, incorrect, illegible or misplaced			X
Markings	Omitted, incorrect, illegible or misplaced			X
Cleanness:	Grease or oil stains			X
	Thread ends not trimmed throughout			X

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TABLE II. End item visual defects (frame)

Examine	Defect	Classification Major	Minor A	Minor B
Welds and welding	Missing, burned through, porous, cracked, fractured or not fused	X		
	Not type specified		X	
	Not smooth and uniform, flash, scale or flux deposits not removed			X
Pin, escutcheon	Pin omitted, not peened, head broken off	X		
	Pin not properly peened; e.g., not tight fit, ragged or sharp edges			X
Pole	Sections will not slide as intended	X		
	Sections require abnormal force to slide			X
	Spindle not tight fit to pole sections but is securely retained in tube			X
	Spindle not securely attached to tubing	X		
Spindle assembly	Ring not joined by welding	X		
	Eye washers not brazed to arm rods	X		
	Wrong size eyes on end of arm	X		
	Eye washer not affixed to rod at proper angle	X		
	NOTE: Eye A and B are brazed at different angles; in addition, eye B has a 20 degree offset.			
	Helical spring has less than 20 coils, is broken, or has insufficient tension to hold arms at proper position	X		
	Any free floating brass washer omitted where required	X		
	End of arm rod not chamfered		X	
	Spider arm not rounded, seamless aluminum tubing	X		
Finish	Omitted; not adherent; e.g., is blistered, peeling or flaking	X		
	Not type specified		X	
	Not dry; i.e., is tacky to touch			X
	Run, drop or sag		X	

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TABLE II. End item visual defects (frame) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Material	Not fabricated from specified material	X		
Construction and workmanship (applicable to all components and assemblies)	Component missing	X		
	Component fractured, split or broken	X		
	Component punctured, dented, twisted, sprung, malformed, bent out of shape or is otherwise impaired	X		
	Any component in wrong location, misplaced, or is not in proper alignment	X		
	Any operation omitted or not properly performed; e.g., hole is not countersunk or ends are not chamfered where required		X	
	Countersink or chamfer is not as specified			X
	Sharp edge, burr or sliver			X
	Casting is not sound; e.g., is porous warped, cracked, fractured	X		

* 4.4.4 End item dimensional examination.

* 4.4.4.1 Tent, observing, triangulation, ground type. The end item shall be in an erected position so that both the inside and outside of the tent can be examined for defects listed in table III. The lot size shall be expressed in units of tents. The sample unit shall be one tent. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects, 15 for major and minor A combined defects and 40 for total (major, minor A, and minor B combined) defects.

* 4.4.4.2 Cover. The end item shall be in a position so that both the inside and outside of the cover can be examined for defects listed in table III. The lot size shall be expressed in units of covers. The sample unit shall be one cover. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units shall be 10 for major defects, 25 for major and minor A combined defects and 100 for total (major, minor A, and minor B combined) defects.

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* 4.4.4.3 Frame. The end items shall be examined for conformance to the dimensions specified on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of frames. The sample unit shall be one frame. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE III. End item dimensional defects (tent or cover)

Examine	Defect	Classification Major	Minor A	Minor B
Overall dimensions: on tent	Smaller than specified dimensions less applicable minus tolerance indicated on drawings	X		
	Larger than specified dimensions and applicable plus tolerances indicated on drawings			X
On covers	Smaller than specified dimensions less applicable minus tolerance indicated on drawings, but not smaller than specified dimensions less twice the applicable minus tolerance			X
	Smaller than specified dimensions less twice the applicable minus tolerance indicated on drawings		X	
	Larger than specified dimensions and applicable plus tolerance indicated on drawings			X
Hems	Hems required to finish less than 3/4 inch in width, less than specified; hems required to finish 3/4 inch or more in width, less than specified by more than 1/4 inch			X
Sod cloths	One or more sod cloths set on wall bottom hem beyond specified location by more than 3/4 inch			X
Location dimensions (unless otherwise specified)	Not within the specified tolerance: -not exceeding twice the applicable tolerance			X
	-exceeding twice the applicable tolerance		X	

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TABLE III. End item dimensional defects (tent or cover) (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Webbing and tie tapes	Finished length less than specified by more than 1/2 inch			
	Turnunder at ends of webbing engaging hardware less than specified:			
	- by more than 1/2 inch but not more than 1 inch			X
	- by more than 1 inch		X	
Grommets (eave)	Set beyond specified location by more than 1/2 inch			X
Reinforcements	Finished dimensions less than specified by more than 1/2 inch			X

* 4.4.5 End item testing. Tents shall be tested as specified in 4.5.1 for compliance with 3.3.12. The lot size shall be expressed in units of tents. The sample unit shall be cause for rejection of the lot.

* 4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Content	Not packaged as specified.

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* 4.5 Methods of inspection.

* 4.5.1 Wicking of sewing thread (cup test). The tent shall be tested in three areas as follows: at the peak, at an intersection of a horizontal and vertical seam such as at an intersection of a drop wall and roof, and on a roof panel seam. Suspend the seam section of the test fabric in the center of a 6 ± 1/8 inch diameter hoop and form a depression with the seamed fabric in the hoop to a depth of 1 ± 1/4 inch. Slowly pour 500 mL of water at 77° ± 4°F into the cupped area (depression) and observe the under surface of the fabric for water penetration. Any wicking of water along the sewing thread, identified by a discoloration or darkening of the thread within 5 minutes after water is poured, shall constitute a failure. Water leakage through the needle holes or between the plies of the lapped seamed fabric shall not be considered a failure.

* 5. PACKAGING.

* 5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each tent top and drop wall shall be folded as shown on Drawing 5-4-487. The spider shall be collapsed and the spider arms brought together. Each sectional upright pole shall be disassembled. The tent lines shall be loosely coiled together and the coil securely tied at each end with cotton twine. The length of the coil shall not exceed 50 inches. The tent line slips, if not attached to the tent lines, shall be strung on cotton twine or stacked and tied together. All the tent components shall be packaged in the tent cover as shown on Drawing 5-4-487.

* 5.1.2 Commercial. Tents shall be preserved in accordance with ASTM D 3951.

* 5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

* 5.2.1 Level A. One complete tent, preserved as specified in 5.1, shall be packed in a snug-fitting crate conforming to type III, style B of MIL-C-52950.

* 5.2.2 Commercial. Tents preserved as specified in 5.1 shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129, or ASTM D 3951, as applicable.

5.3.1 Additional marking. The following shall be stenciled or imprinted on each container in characters not less than 3/4 inch high:

"SEE ERECTION INSTRUCTION SEWED TO INSIDE OF TENT COVER"

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6. NOTES

6.1 Intended use. The tent is intended for use as a shelter for personnel and instruments, in survey and mapping operations by the Inter-American Geodetic Survey (IAGS), Caribbean Defense Command, the Army Map Service, and by astronomic and missile site survey teams. A cover is provided to protect the tent during transportation.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required (see 3.1, 4.3 and 6.4).
c. Applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 Sample. For access to a standard shade sample of thread, address the contracting activity issuing the invitation for bids.

* 6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.6 Other components. The tent covered by this document consists of the following components:

a. Main wall; Tent, Observing, Triangulation, Ground Type.
b. Top and drop wall; Tent, Observing, Triangulation, Ground Type.
c. Cover, for Tent, Observing (Astronomic, Ground Type, Magnetic or Tower Type).
d. Frame; Tent, Observing, Triangulation, Ground Type.
e. Line, tent (eave guy), type XXI, 13 feet long (6 each).
f. Line, tent (footstop), type I, 19 inches long (15 each).
g. Slips, tent line (6 each).

In addition to the components listed above, the following are required to erect the Tent, Observing, Triangulation, Ground Type, Complete with Frame.

21 - Pins, Tent Aluminum, 9-inches.

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6.7 Seam retreatment. A satisfactory method of applying the textile compound for the repair of leaking seams is as follows:

a. A suitable volume of textile compound conforming to type I, class 1 of TT-P-595, (55 gallons) shall be thoroughly stirred and diluted with an equal volume of petroleum solvent (Stoddard solvent or equivalent). Thoroughly stir the mixture again before use.

b. Prior to application of the compound, the tents must be dry. Special attention must be taken to insure adequate ventilation and that proper fire preventive measures are taken since the compound is flammable.

c. Two coats of the textile compound shall then be applied to the seams (on the outside of the tent only). A 4-inch wide stiff brush is recommended for use. The first coat shall be air dried for a minimum of 4 hours prior to application of the second coat. Elapsed time between coats may be reduced by providing a supply of hot air to evaporate the solvent more quickly. In this manner, a second coat may be applied as soon as the first coat is dry to the touch. After the application of the second coat, the treated tent shall be allowed to air dry for a period of at least 24 hours.

6.7.1 Seams to be treated. Seams which have failed the test specified in 4.5.1 shall be treated as follows:

a. Failure at roof panel lap seams	- treat all roof panel lap seams.
b. Failure at eave (except "W" stitching)	
1. Stitching joining roof and wall	- treat all stitching on fabric eave.
2. Tape stitching	- treat all tape stitching on roof.
c. Failure at peak	
1. Stitching attaching peak reinforcement	- treat all stitching on roof reinforcements.
2. Tape stitching	- treat all tape stitching on roof

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* 6.8 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG NO. 2882, relative to camouflage requirements for tents, shelters and subsidiary components. When amendment, revisions, or cancellation of this document is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

* 6.9 Subject term (key word) listing.

Shelter
Tent, Observing, Triangulation

* 6.10 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-40001E	TENT, OBSERVING, TRIANGULATION, GROUND TYPE, COMPLETE WITH FRAME

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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